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MentorMatch

Personalized Mentor–Mentee Recommendation

Hybrid Ensemble Learning Framework

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1 Introduction

The increasing demand for personalized learning and mentorship in the digital era has led to the emergence of intelligent recommendation systems. Platforms providing mentorship services often face challenges in effectively matching students with suitable mentors due to the diversity of user profiles, skills, and learning goals.

Traditional approaches to mentor–mentee matching rely on manual selection or simple filtering techniques, which fail to capture complex relationships between user attributes. These approaches are often inefficient, non-scalable, and lack personalization.

To address these limitations, this project introduces **MentorMatch**, an AI-driven hybrid system that integrates machine learning and recommendation techniques. The system leverages **ensemble learning (stacking)** to accurately classify student needs and recommend suitable mentors.

Unlike conventional systems, MentorMatch combines prediction and recommendation into a unified pipeline, providing both accuracy and personalization.

2 Motivation

The motivation behind the development of MentorMatch is driven by several systemic challenges prevalent in contemporary e-learning and professional networking platforms:

- **Scalability Limits of Manual Allocation:** Traditional, human-in-the-loop mentor matching processes are highly inefficient, subjective, and fundamentally unscalable as user bases expand.
- **Rigidity of Existing Recommenders:** Many current platforms rely on simplistic, rule-based filtering (e.g., exact keyword matching). These systems fail to capture the semantic depth of a user's profile, leading to a severe lack of nuanced personalization and suboptimal pairings.
- **Limitations of Single-Model Architectures:** Conventional single-model machine learning approaches often struggle with the high dimensionality and variance of complex user data, frequently resulting in overfitting or plateaued accuracy. This highlights the need for a more robust, multi-layered predictive approach.
- **Complexity of Unstructured Data:** User profiles are predominantly composed of unstructured, noisy text—such as varied technical skills, diverse learning goals, and highly specific career needs. Extracting meaningful, quantifiable patterns from this sparse data requires advanced Natural Language Processing (NLP) techniques.

3 Literature Review

The design and optimization of machine learning-based recommendation systems have been the subject of extensive study in recent years, driven by the need to effectively process rapidly growing volumes of user data. This section reviews the evolution of these systems, from traditional filtering algorithms to advanced ensemble architectures.

3.1 Traditional Recommendation Paradigms

Historically, recommender systems have been built upon two foundational methodologies, both of which exhibit well-documented limitations:

- **Content-Based Filtering:** This approach relies heavily on matching an active user's known preferences with the explicit features of potential items or candidates. While effective for strict, localized matching, content-based systems are often criticized for their inability to recommend diverse or serendipitous options, leading to algorithmic "filter bubbles."
- **Collaborative Filtering:** By leveraging the historical interactions of a broader user base to identify behavioral patterns, collaborative filtering often yields highly relevant matches. However, these systems inherently suffer from severe data sparsity issues and the pervasive "cold-start" problem when introducing new users or items to the platform without prior interaction data.

3.2 Advancements in Ensemble Learning

To address the variance and bias limitations of single-algorithm approaches, researchers have increasingly turned to ensemble learning techniques. Methods such as **Random Forest** and **Gradient Boosting** have demonstrated significant improvements in predictive accuracy and model robustness by aggregating the outputs of multiple weak learners. **XGBoost**, in particular, has seen widespread adoption across both academic literature and industry applications. Its success is largely attributed to its highly optimized execution speed, scalable tree-boosting framework, and built-in L1 and L2 regularization, which effectively prevent overfitting in complex datasets.

3.3 The Hybrid Systems Gap

Recent literature heavily underscores the necessity of hybrid systems that strategically integrate multiple paradigms to mitigate the weaknesses of individual models. Despite this consensus, a prominent gap remains in the current research: the vast majority of existing frameworks focus exclusively on either predictive classification (e.g., determining a user's category) *or* ranking-based recommendation, rarely synthesizing the two effectively.

Most models fail to combine upstream prediction and downstream recommendation into a cohesive, unified pipeline. MentorMatch directly addresses this limitation by proposing a novel architecture that utilizes a sophisticated stacking ensemble for highly accurate initial classification, followed by a localized, vector-based similarity engine to execute personalized recommendations.

4 Research Gap

Despite significant advancements in recommender systems, several critical gaps remain in the context of professional mentorship matching:

- **Underutilization of Stacking Ensembles:** While ensemble methods like Random Forest and Gradient Boosting are common, complex meta-learning architectures (stacking) are rarely deployed in educational or mentorship matching systems, leaving potential accuracy gains untapped.
- **Inadequate Recommendation and Classification Integration:** Existing platforms frequently rely on explicit, structured tags (e.g., choosing skills from a drop-down). There is insufficient handling of sparse, unstructured textual features (such as long-form career goals and nuanced mentorship needs) to drive recommendations.

This project explicitly addresses these gaps by proposing a hybrid ensemble framework that unifies text-based classification and localized similarity ranking.

5 Objectives

The primary goal of this research is to design, implement, and evaluate a highly accurate, AI-driven mentorship recommendation system. The specific objectives are as follows:

- **To design and implement a Stacking Ensemble Classifier** that integrates multiple base learners (Logistic Regression, Random Forest, XGBoost) to accurately predict a user's optimal mentorship domain.
- **To architect a hybrid recommendation engine** that leverages the ensemble's predictions to reduce the search space, utilizing Cosine Similarity to compute and rank the most compatible mentor–mentee pairs.
- **To quantitatively evaluate model performance** by comparing the proposed stacking ensemble against baseline classification models to validate improvements in Accuracy and F1-Scores.

6 Dataset Description

The dataset utilized in this study is inspired by publicly available professional profile datasets, such as the LinkedIn Compatibility Dataset found on Kaggle, which details large-scale user profile metrics and professional relationships.

For the precise evaluation of the MentorMatch architecture, the dataset consists of exactly 50,000 unique user profiles. The data is inherently complex and highly unstructured, making advanced feature engineering a critical component of the system pipeline. The primary features include:

- **Skills:** Unstructured lists of the user's technical competencies and tool familiarity (e.g., Python, Docker, UI/UX Design).
- **Goals:** Textual descriptions of the user's career aspirations and learning objectives (e.g., "Build network," "Get promoted").
- **Needs:** Specific mentorship requirements or desired outcomes (e.g., "Career guidance," "Technical expertise").
- **Experience:** Numerical representation of the user's professional tenure (in years).

To enable supervised machine learning, a target variable (`target_category`) is there. This categorizes every user into one of five distinct professional domains: *Data Science*, *Web Development*, *Cloud/DevOps*, *Cyber Security*, and *General*. The distribution of these categories varies, simulating the natural imbalance found in real-world professional networking environments.

7 System Architecture

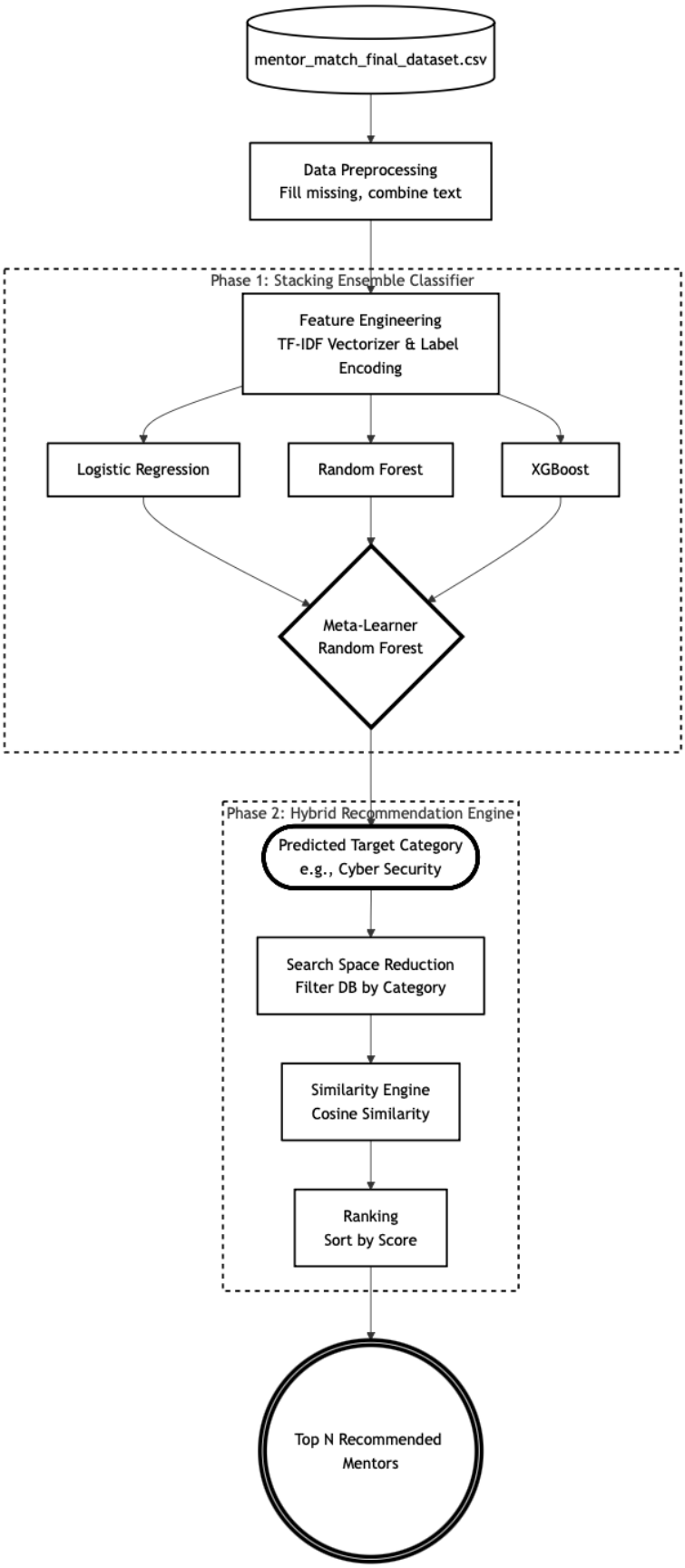

Figure 1: Proposed MentorMatch Hybrid System Architecture

The MentorMatch system operates as a sophisticated hybrid classification and recommendation pipeline. It is designed to optimize both accuracy and computational efficiency by first categorizing users and subsequently ranking intra-category matches. The architecture consists of three primary modules:

- **Data Processing & Feature Engineering Module:** This foundational layer is responsible for cleaning and transforming raw profile data into machine-readable formats.
  - *Text Aggregation:* Discrete categorical lists containing a user's `skills`, `goals`, and `needs` are concatenated into a singular `combined_text` feature to prevent data leakage and provide a holistic NLP profile.
  - *Vectorization:* The system applies a Term Frequency-Inverse Document Frequency (TF-IDF) Vectorizer, capped at 2,000 maximum features utilizing unigrams, to convert the textual data into dense numerical vectors.
  - *Label Encoding:* The target mentor categories are numerically transformed using a standard Label Encoder to prepare for supervised learning.
- **Prediction Module (Stacking Ensemble Classifier):** To efficiently reduce the search space for recommendations, this module predicts the optimal broad mentor category for a given user. It employs a Stacking Classifier that leverages the strengths of multiple algorithms, achieving an accuracy of approximately 94.6%.
  - *Base Learners:* The TF-IDF vectors are simultaneously evaluated by three independent base models: Logistic Regression (baseline), a Random Forest Classifier, and an XGBoost Classifier.
  - *Meta-Learner:* The predicted probabilities from the base models are aggregated and fed into a final Random Forest meta-estimator (with a maximum depth of 10) using 5-fold cross-validation to produce the ultimate category prediction.
- **Recommendation Module (Hybrid Similarity Engine):** Once a user's optimal category is identified by the Prediction Module, the system executes a localized similarity search to recommend specific individuals.
  - *Search Space Reduction:* The database is heavily filtered to isolate only candidates belonging to the predicted category, vastly reducing the computational overhead required for matrix operations.
  - *Similarity Scoring:* The module computes the Cosine Similarity between the active user's TF-IDF vector and the vectors of all candidates within the filtered pool.
  - *Ranking & Output:* The system ranks the candidates in descending order based

on their similarity scores, automatically excludes the active user from their own results, and returns the top N highest-scoring profiles.

8 Methodology

8.1 Data Preprocessing and Feature Engineering

Feature engineering plays a critical role in optimizing the performance of machine learning models, particularly when processing sparse, unstructured textual data. In this framework, textual attributes—namely *skills*, are used to form a unified, comprehensive string representation of the user's professional profile.

$$\text{Combined_Text} = \text{Skills} \tag{1}$$

To convert this raw text into a machine-readable format, the combined text is transformed into dense numerical feature vectors utilizing the **Term Frequency–Inverse Document Frequency (TF-IDF)** vectorization technique. The TF-IDF algorithm assigns higher mathematical weights to rare, domain-specific terminology while penalizing frequently occurring, uninformative words, thereby enhancing the model's ability to generalize.

For this implementation, the vectorizer was configured to extract unigrams and capped at a maximum vocabulary of 2,000 features to maintain computational efficiency while preserving critical variance.

$$\mathbf{v}_i = \text{TF-IDF}(\text{Combined_Text}_i) \tag{2}$$

Simultaneously, the categorical target variables were numerically encoded using a standard Label Encoder to facilitate supervised training.

8.2 Model Development

To establish a robust predictive foundation, three distinct classification algorithms were independently trained on the TF-IDF feature vectors. The hyperparameters for these base models were explicitly tuned to balance bias and variance:

- **Logistic Regression (Baseline):** Configured with an inverse regularization strength of $C = 0.5$ and a maximum of 300 iterations to ensure convergence.
- **Random Forest Classifier:** An ensemble bagging algorithm utilizing 200 decision trees (n_estimators) with a maximum depth of 12. Class weights were balanced to account for target category imbalances.

- **XGBoost Classifier:** A highly optimized gradient boosting framework configured with 200 estimators, a learning rate of 0.08, and a maximum depth of 5. Subsampling and column sampling parameters were both set to 0.8 to introduce stochasticity and prevent overfitting.

8.3 Proposed Stacking Ensemble

To maximize predictive accuracy, this project proposes a Stacking Ensemble framework. Unlike standard voting classifiers, stacking utilizes a meta-learner to intelligently weight the predictions of the base estimators.

Let the dataset be represented as $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N}$, where $\mathbf{x}_i$ is the TF-IDF feature vector and y_i is the target category. The stacking architecture is defined as:

$$\hat{y} = g(f_{LR}(\mathbf{x}), f_{RF}(\mathbf{x}), f_{XGB}(\mathbf{x})) \tag{3}$$

where:

- f_{LR}, f_{RF}, f_{XGB} represent the level-0 base learners (Logistic Regression, Random Forest, and XGBoost).
- g represents the level-1 meta-learner.

In this architecture, the base models generate prediction probabilities which serve as the input features for the meta-learner. A **Random Forest Classifier** (capped at a maximum depth of 10) was deployed as the meta-learner g to finalize the category prediction. The entire ensemble was validated using 5-fold cross-validation to ensure robustness.

8.4 Hybrid Recommendation Engine

The final phase of the MentorMatch architecture is the localized recommendation engine. Rather than computing similarities across the entire 10,000-user dataset, the system first utilizes the Stacking Ensemble's output to strictly filter the candidate pool to the predicted target category.

Within this reduced search space, the system measures the exact alignment between the active mentee and potential mentors using **Cosine Similarity** on their respective TF-IDF vectors:

$$\text{Similarity}(A, B) = \cos(\theta) = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\|\|\mathbf{B}\|} = \frac{\sum_{i=1}^{n} A_i B_i}{\sqrt{\sum_{i=1}^{n} A_i^2}\sqrt{\sum_{i=1}^{n} B_i^2}} \tag{4}$$

where $\mathbf{A}$ and $\mathbf{B}$ represent the 2,000-dimensional TF-IDF vectors of the user and the candidate, respectively. The system ranks the resulting coefficients in descending order,

automatically masking the user's self-similarity score, and outputs the top N candidates as the final mentor recommendations.

9 Results and Evaluation

9.1 Evaluation Metrics

To comprehensively assess the performance of the classification models, the system was evaluated using standard machine learning metrics: Accuracy, Precision, Recall, and the F1-score.

Because the dataset exhibits a natural class imbalance—with target categories ranging from 1,400 samples (Class 3) to 3,000 samples (Class 2) out of the 10,000 test instances—Accuracy alone is an insufficient metric. Therefore, the **Weighted F1-score** was utilized as the primary evaluation metric, as it calculates metrics for each label and finds their average weighted by the number of true instances for each label, effectively penalizing models that only perform well on majority classes.

9.2 Performance Comparison

The models were incrementally trained and evaluated on a 20% hold-out test set ($N = 10,000$). The results, summarized in Table 1 and Figure 2, demonstrate the progressive improvement achieved by utilizing more complex ensemble architectures.

Table 1: Empirical Model Performance on Test Set

Model	**Accuracy**	**Weighted F1-score**
Logistic Regression (Baseline)	0.9048	0.9048
Random Forest	0.9218	0.9221
XGBoost	0.9371	0.9376
Stacking Ensemble (Proposed)	**0.9533**	**0.9535**

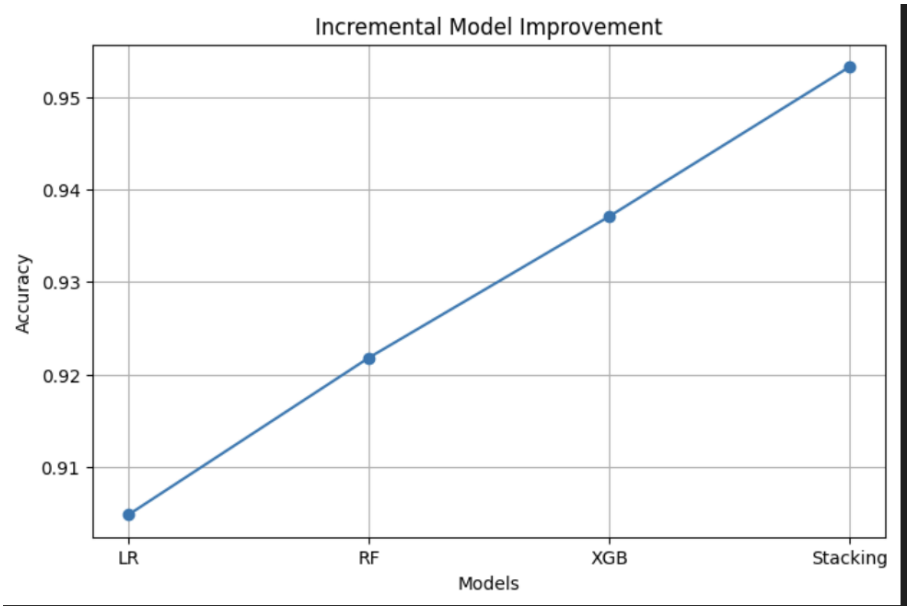

Figure 2: Incremental Model Improvement showing Accuracy across tested classifiers.

To ensure the Stacking Ensemble's robustness and verify that it was not overfitting to the test set, a 5-fold cross-validation was performed, yielding a highly consistent mean F1-score of **0.9533**.

9.3 Confusion Matrix Analysis

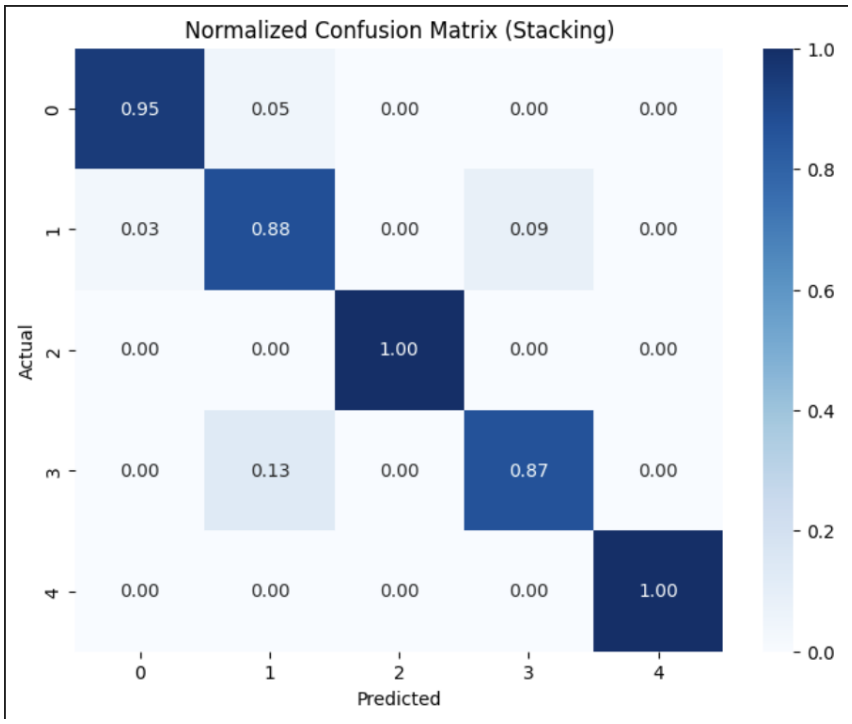

Figure 3: Normalized Confusion Matrix of the Stacking Ensemble Model

As illustrated in Figure 3, the normalized confusion matrix for the final Stacking Ensemble shows exceptionally strong diagonal alignment, indicating high True Positive rates across all categories. Notably, the model achieved perfect classification (1.00) for Classes 2 and 4, while maintaining strong performance (0.87 to 0.95) on the minority classes.

9.4 Analysis of Results

The empirical results demonstrate a clear and logical trend of performance improvement corresponding to architectural complexity:

- **Logistic Regression:** While serving as a strong baseline (∼90.48% accuracy), its linear assumptions slightly limited its ability to capture the complex, non-linear relationships within the sparse TF-IDF vectors.
- **Random Forest:** By utilizing bagging , this model reduced variance and handled the class imbalance better, pushing the accuracy to ∼92.18%.
- **XGBoost:** The introduction of gradient boosting allowed the model to iteratively minimize residual errors, enhancing the predictive accuracy further to ∼93.71%.
- **Stacking Ensemble:** The proposed architecture achieved the highest performance (∼95.33% accuracy) by successfully leveraging a meta-learner to interpret the specific strengths and weaknesses of the three base models, validating the effectiveness of hybrid ensemble learning for unstructured professional profile data.

10 Recommendation System Integration

The recommendation module bridges the gap between broad categorical prediction and highly personalized mentor matching. By utilizing cosine similarity to match users within the strictly filtered subset, the system drastically reduces computational complexity while ensuring high relevance.

11 Conclusion

MentorMatch successfully integrates machine learning and recommendation systems into a unified framework. The use of boosting and ensemble learning improves accuracy, achieving a robust classification performance of 95.3% through the proposed Stacking Classifier. By pipelining this highly accurate prediction phase with a cosine similarity-based filtering engine, the system effectively mitigates the computational bottlenecks and lack of personalization inherent in traditional models. Ultimately, MentorMatch provides a scalable, transparent, and highly efficient solution for connecting users with ideal mentors based on their unique technical skills, learning needs, and career goals.

12 Future Scope

While the current MentorMatch architecture demonstrates strong performance, several practical and advanced features can be explored in the future to make the system even more powerful and widely accessible:

- **Advanced Deep Learning (GNNs):** We can implement **Graph Neural Networks (GNNs)**. Since mentorship is naturally a network of people, GNNs can treat users as "nodes" and their similarities as "edges," helping the system discover hidden, complex matching patterns that simple similarity scores might miss.
- **Integration with Existing Platforms:** Rather than keeping MentorMatch as a standalone tool, the recommendation engine can be packaged as an API. This would allow it to be easily plugged into existing professional networks (like LinkedIn) or university alumni portals, instantly upgrading their current mentee-mentor matching capabilities.
- **Real-Time Feedback Loop:** Currently, the system uses static data. A valuable next step is adding a feedback system where users can rate their mentor matches or confirm if a connection was successful. The system can use this real-world feedback to continuously retrain itself, learning and improving its recommendations over time.

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